



INFORMATION SYSTEMS

CURRICULUM BOOK 2025

Study Program of Information Systems
Faculty of Computer Science
President University

Intelligent Enterprise Systems • Data Science for Business Analytics • AI-Powered Enterprise Automation

Curriculum Development Team

2025

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Preface

The Information Systems Study Program Curriculum 2025 is prepared as a strategic academic document to guide the implementation of teaching, learning, assessment, and graduate competency development within the program. This curriculum is designed to respond to the rapid transformation of information systems practices in industry, business, government, and society, particularly in relation to intelligent enterprise systems, data-driven business analytics, artificial intelligence, automation, digital transformation, and work-integrated learning.

The development of this curriculum reflects the commitment of the Information Systems Study Program to implement Outcome-Based Education (OBE), in which the learning process is directed toward the achievement of clearly defined graduate profiles and Program Learning Outcomes (PLOs). The curriculum is also structured to strengthen the connection between academic learning and professional practice through project-based learning, concentration-based bootcamps, career streams, and capstone projects that address real organizational and industrial problems.

This curriculum book presents the rationale, objectives, vision and mission alignment, graduate profiles, Program Learning Outcomes, curriculum structure, course mapping, course descriptions, and related professional pathways. It is expected to serve as a reference for lecturers, students, academic managers, curriculum reviewers, industry partners, and other stakeholders in understanding the direction and implementation of the Information Systems Study Program Curriculum 2025.

The curriculum emphasizes the development of graduates who are not only technically competent, but also professional, ethical, communicative, adaptive, collaborative, entrepreneurial, and capable of lifelong learning. Through the integration of core Information Systems knowledge, enterprise applications, business analytics, AI-powered automation, and practical industry exposure, the program aims to prepare graduates who can contribute meaningfully to digital transformation and innovation in various sectors.

We would like to express our appreciation to all parties who have contributed to the preparation of this curriculum, including faculty members, university leaders, students, alumni, industry partners, professional communities, and other stakeholders. Their valuable input has helped ensure that this curriculum remains relevant, practical, and aligned with current and future demands.

We hope that this curriculum book can become a useful guide for the continuous improvement of the Information Systems Study Program and support the development of graduates who are ready to compete professionally, contribute to industry and society, and continue learning in an ever-changing digital world.

Curriculum Development Team

Information Systems Study Program
2025

1. Background and Rationale

The 2025 curriculum is developed to respond to rapid changes in Information Systems education, digital transformation, artificial intelligence, data-driven decision-making, enterprise systems, and automation. Information Systems graduates are increasingly expected to connect business problems with computing-based solutions, manage enterprise-level systems, communicate with stakeholders, and deliver technology solutions that are ethical, reliable, and relevant to organizational needs.

The curriculum is designed as a 3.5-year curriculum consisting of 9 semesters and 144 credits. It combines fundamental computing and Information Systems knowledge, applied system development, concentration-based bootcamp learning, work-integrated learning, and a capstone project. The curriculum started in 2025 to ensure stronger alignment between graduate competencies and the needs of industry, enterprises, government, and society.

The curriculum also responds to labour market changes. The World Economic Forum Future of Jobs 2025 reports that global labour markets are being reshaped by technological development and that employers expect 39% of key skills to change by 2030. It also identifies AI and big data, networks and cybersecurity, technological literacy, creative thinking, leadership, resilience, and lifelong learning as important rising skills. Therefore, the curriculum emphasizes not only technical capability but also communication, ethics, teamwork, entrepreneurship, and continuous learning.

2. Curriculum Objectives

- a. To implement the 2025 curriculum as a 3.5-year, 9-semester curriculum structure that is more responsive to industry needs and digital transformation.
- b. To strengthen graduate competencies in intelligent enterprise systems, data science for business analytics, and AI-powered enterprise automation.
- c. To align graduate profiles, Program Learning Outcomes (PLO), curriculum structure, course objectives, and assessment with an Outcome-Based Education approach.
- d. To provide work-integrated learning, concentration-based bootcamp learning, and capstone project experiences that require students to solve authentic organizational problems.
- e. To prepare graduates who are professional, ethical, communicative, collaborative, entrepreneurial, and able to learn continuously in response to technological change.

3. Vision and Mission

3.1 Vision

To become a leading Information Systems study program in Indonesia that excels globally in intelligent enterprise systems, data-driven business solutions, and AI-powered enterprise automation to support digital transformation, industry innovation, and sustainable societal development.

3.2 Mission

1. To provide high-quality education and learning in Information Systems that integrates enterprise systems, business analytics, artificial intelligence, automation, and emerging digital technologies.
2. To develop graduates who are professional, ethical, adaptive, and capable of designing, implementing, and managing Information Systems solutions for business, industry, government, and society.
3. To conduct innovative and applied research in Information Systems, particularly in intelligent enterprise systems, data science for business analytics, AI-powered automation, and digital transformation.
4. To carry out community service and empowerment activities by applying Information Systems knowledge and technology to solve practical problems in society, organizations, and industry.
5. To strengthen collaboration with industries, enterprises, government institutions, professional associations, and global partners in education, research, innovation, and work-integrated learning.
6. To foster technopreneurship, lifelong learning, leadership, and interdisciplinary collaboration among students and academic communities.

3.3 Alignment between Vision-Mission and Curriculum Objectives

The alignment table below shows how the renewed vision and mission are translated into concrete curriculum objectives. This alignment ensures that strategic statements are implemented through courses, learning experiences, professional preparation, and graduate outcomes.

Vision/Mission Emphasis	Related Curriculum Objective
Intelligent enterprise systems	Strengthen enterprise-level Information Systems competencies and prepare graduates for enterprise systems implementation and transformation.
Data-driven business solutions	Develop data science, business analytics, business intelligence, and data-informed decision-making competencies.
AI-powered enterprise automation	Prepare students for AI-enabled process automation, RPA, workflow automation, document processing, and process mining.
Industry innovation and work-integrated learning	Strengthen internship, career stream, bootcamp project, and capstone project experiences.
Technopreneurship	Support entrepreneurial ability to design IS-based business or start-up initiatives.
Lifelong learning and knowledge transfer	Prepare graduates to adapt to continuous technological change and contribute to knowledge development.

4. Graduate Profile, Professional Orientation, and Program Learning Outcomes

4.1 Graduate Profile

The graduate profiles describe the intended identity and professional direction of Information Systems graduates. These profiles become the foundation for defining the Program Learning Outcomes and for determining the types of professions targeted by the curriculum.

Code	Graduate Profile	Description	Related PLO
GP-1	Information Systems Professional and Enterprise Change Leader	Graduates have strong professional skills in Information Systems and are able to lead enterprise-level transformation through the effective use of Information Systems technology.	PLO-1, PLO-2, PLO-4, PLO-5, PLO-6
GP-2	Technopreneur	Graduates have a technopreneurial mindset and are able to build and lead independent business initiatives by combining Information Systems technologies with interdisciplinary knowledge.	PLO-2, PLO-5, PLO-6, PLO-7
GP-3	Knowledge Transfer, Scientific Development, and Lifelong Learner	Graduates are able to transfer knowledge and expertise, contribute to the development of Information Systems through innovative, creative, and applied works, and engage in lifelong learning.	PLO-3, PLO-4, PLO-8

4.2 Program Learning Outcomes (PLO/CPL)

The Program Learning Outcomes define the competencies that students are expected to demonstrate by the time they graduate. Each PLO is later supported by selected courses, concentration pathways, work-integrated learning, and capstone activities.

Code	Short Label	Program Learning Outcome
PLO-1	Complex Problem Analysis	Ability to identify and analyze complex computing problems using computing principles and relevant interdisciplinary knowledge to produce creative, innovative, and current solutions.
PLO-2	Computing Solution Design and Implementation	Ability to design computing-based solutions, implement them using modern tools and techniques, and evaluate the resulting solutions in a specific domain.
PLO-3	Communication	Ability to communicate effectively and appropriately, both orally and in writing, across various domains.
PLO-4	Professional, Ethical, and Data-Informed Practice	Ability to act responsibly and professionally and to make judgments based on data and information while considering legal and ethical norms in computing practice.
PLO-5	Leadership and Teamwork	Ability to function effectively and efficiently as a leader or team member in Information Systems-related activities and other collaborative activities.
PLO-6	Information Systems Management and Business/Industrial Application	Ability to support the idea, delivery, use, and management of Information Systems in business, industrial, and other organizational environments.
PLO-7	Technopreneurship	Entrepreneurial attitude and ability to design an independent business or start-up related to Information Systems.

PLO-8	Lifelong Learning and Knowledge Transfer	Ability to learn independently throughout life, create knowledge, and transfer up-to-date knowledge.
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4.3 Graduate Profile to PLO Logic

The graduate profiles are translated into Program Learning Outcomes so that each profile is measurable through the curriculum. GP-1 is supported by analytical, design, ethical, leadership, and enterprise management outcomes. GP-2 is supported by solution design, leadership, enterprise understanding, and technopreneurship outcomes. GP-3 is supported by communication, ethics, and lifelong learning outcomes.

5. Related Professions, KKNi Linkage, and Industry Relevance

5.1 Basis for Profession Selection

The profession list is derived from the previous profession mapping document, which links several Information Systems-related occupations to the National ICT Occupation Map and KKNi qualification levels. The existing profession list includes Programmer Supervisor, System Analyst, ERP Officer, Data Engineer Supervisor, System Development Manager, and Database Administrator. This version expands the profession list to reflect the updated 2025 curriculum concentrations, especially AI-Powered Enterprise Automation, which was not yet fully covered in the previous profession list.

For official occupation references, the curriculum uses occupations with available codes from the submitted profession mapping. For new and emerging roles, especially in AI-powered automation, the roles are presented as aligned or emerging professions. They are mapped to KKNi-equivalent levels based on expected graduate competence, responsibility, and complexity, but their official occupation codes should be verified later against the latest National ICT Occupation Map when available.

5.2 Existing Occupations Linked to KKNi and the Curriculum

The following table presents the existing occupations that are already linked to the national ICT occupation map and KKNi qualification levels. These professions represent the core career orientation of the Information Systems curriculum and provide a formal reference for graduate employability.

Profession	Occupation Code	KKNi Level	Related Curriculum Area	Scope of Work
Programmer Supervisor / Information Systems Full-Stack Developer	TIK.DEV0601	KKNi Level 6	All; Capstone; system development foundation	Develops and supervises end-to-end information system solutions, including front-end, back-end, database, integration, quality, and documentation.
System Analyst	TIK.DEV0613	KKNi Level 6	All; system analysis, business process, capstone	Analyzes organizational needs and translates them into information system requirements, models, and solution designs.
ERP Officer / Enterprise Systems Officer	TIK.DEV0618	KKNi Level 6	Intelligent Enterprise Systems	Implements, configures, integrates, and maintains ERP or enterprise systems used to manage integrated business processes.
Data Engineer Supervisor	TIK.DSC0603	KKNi Level 6	Data Science for Business Analytics	Designs and supervises data pipelines, databases, integration, and data processing that support analytics and decision-making.
System Development Manager	TIK.DEV0701	KKNi Level 7	Capstone, WIL, project management	Plans and coordinates system development processes so that information systems meet organizational needs and create business value.
Database Administrator	TIK.DSC0604	KKNi Level 6	Database, BI, data science	Manages, secures, maintains, and optimizes databases so that organizational data remains reliable, accessible, and protected.

5.3 Additional and Emerging Professions for the 2025 Curriculum

In addition to the formally mapped occupations, the 2025 curriculum also prepares students for emerging digital professions. These roles are especially relevant to the new concentration areas, including AI-powered automation, enterprise workflow integration, intelligent document processing, and data-driven business transformation.

Profession	Status/Code	KKNI Alignment	Related Curriculum Area	Scope of Work
Business Intelligence Analyst	Emerging / aligned occupation	KKNI Level 6	Data Science for Business Analytics; BI and Analytics	Designs dashboards, reports, and KPI analyses to support managerial decisions.
Data Analyst / Data Scientist for Business	Emerging / aligned occupation	KKNI Level 6	Data Science for Business Analytics	Uses statistics, machine learning, and business understanding to generate insights and predictive models.
Data Visualization and Reporting Specialist	Emerging / aligned occupation	KKNI Level 6	Data Science for Business Analytics	Transforms data into visual stories, dashboards, and management reports for decision-making.
Intelligent Enterprise Systems Analyst	Emerging / aligned occupation	KKNI Level 6	Intelligent Enterprise Systems	Analyzes, configures, and improves enterprise systems by combining ERP, analytics, and AI-enabled functions.
Enterprise Systems Consultant	Emerging / aligned occupation	KKNI Level 7	Intelligent Enterprise Systems; WIL; Capstone	Leads advisory and implementation projects for enterprise system transformation and process integration.
RPA Developer / Automation Analyst	Emerging / aligned occupation	KKNI Level 6	AI-Powered Enterprise Automation	Designs, develops, tests, and monitors automation workflows for repetitive business processes.
AI Workflow and Agent Designer	Emerging / aligned occupation	KKNI Level 6	AI-Powered Enterprise Automation	Designs human-in-the-loop workflows and AI-agent-supported processes with controls, validation, and integration.
Low-Code Application and API Integration Developer	Emerging / aligned occupation	KKNI Level 6	AI-Powered Enterprise Automation	Builds rapid enterprise applications and integrates systems using low-code platforms, APIs, and workflow connectors.
Intelligent Document Processing Specialist	Emerging / aligned occupation	KKNI Level 6	AI-Powered Enterprise Automation	Applies document AI, OCR, information extraction, and validation mechanisms to automate document-heavy processes.
Process Mining Analyst	Emerging / aligned occupation	KKNI Level 6	AI-Powered Enterprise Automation	Analyzes event logs to discover, monitor, and improve business processes and automation opportunities.
Enterprise Automation Consultant	Emerging / aligned occupation	KKNI Level 7	AI-Powered Enterprise Automation; WIL; Capstone	Leads enterprise automation initiatives that combine process analysis, RPA, AI agents, data integration, and governance.

5.4 Relevance of Professions to Curriculum Objectives

The profession-to-objective mapping below explains why each targeted profession is relevant to the curriculum design. It shows how the curriculum objectives support the development of technical competence, business understanding, professional responsibility, leadership, and lifelong learning.

Curriculum Objective	Related Professions	Rationale
Strengthen intelligent enterprise systems competencies	ERP Officer, Intelligent Enterprise Systems Analyst, Enterprise Systems Consultant	The curriculum includes enterprise information systems, intelligent enterprise systems, supply chain, sales, manufacturing, finance, and enterprise systems project courses. These support roles that configure, integrate, and improve enterprise-level systems.
Develop data science and business analytics competencies	Data Engineer Supervisor, Database Administrator, Business Intelligence Analyst, Data Analyst / Data Scientist, Data Visualization Specialist	The curriculum includes database, statistics, AI, BI, data science programming, machine learning, data mining, visualization, and data science project courses. These support data-driven analysis, reporting, and decision-making roles.
Prepare graduates for AI-powered enterprise automation	RPA Developer, Automation Analyst, AI Workflow and Agent Designer, Low-Code Application Developer, IDP Specialist, Process Mining Analyst, Enterprise Automation Consultant	The new concentration directly covers process automation, RPA, AI agents, workflow automation, low-code applications, API integration, intelligent document processing, and process mining.
Strengthen authentic professional experience	Programmer Supervisor, System Analyst, System Development Manager, Enterprise Automation Consultant	Work-integrated learning and capstone project require students to solve real organizational problems, work in teams, communicate with stakeholders, and deliver usable artifacts.
Support technopreneurship and lifelong learning	Technopreneur, Product Founder, Automation Consultant, Digital Transformation Consultant	The curriculum includes entrepreneurship-oriented learning, project-based work, emerging technology exposure, and continuous learning outcomes.

5.5 Why These Professions Are Relevant to Current and Future Demand

The selected professions are relevant because organizations increasingly need graduates who can combine business understanding, computing, data, AI, automation, enterprise systems, and communication. The Future of Jobs 2025 highlights rapid growth in technology-driven roles such as big data specialists, AI and machine learning specialists, and software developers, while also emphasizing that technological skills, AI and big data, networks and cybersecurity, leadership, and lifelong learning are among the skills rising in importance. The OECD also reports that AI changes the tasks and skills required across many jobs, including management, business, cognitive, and digital skills. Therefore, the curriculum is designed not only to prepare students for existing occupational categories but also to equip them for emerging roles created by AI adoption and enterprise automation.

In the local context, the program is positioned near industrial areas and enterprise environments, so graduates are expected to be able to contribute to digital transformation projects, data-driven management, process automation, enterprise systems implementation, and capstone-based organizational problem solving. This makes the profession orientation consistent with the curriculum objectives and graduate profiles.

6. Curriculum Structure and Course-PLO Mapping

6.1 Curriculum Structure Summary

The curriculum summary provides an overview of the 9-semester structure and total credit distribution. It helps readers understand the overall learning path from foundational Information Systems competencies to concentration-based bootcamp learning, work-integrated learning, and capstone implementation.

Semester	Curriculum Block	Credits
1	Fundamental of Information System	20
2	System Development and Automation	19
3	Information System Social Project	9
4	Applied Information System	21
5	IS Concentration / Bootcamp	18
6	IS Concentration / Bootcamp Project	9
7	Career Stream in Information System	20
8	Career Stream and Pre Capstone Project	22
9	Capstone Project	6

Total credits: 144 credits.

6.2 Detailed Curriculum Structure

Semester 1

Semester 1 introduces fundamental Information Systems, computing, business, and organizational concepts. The courses in this semester build the initial analytical, technical, communication, and professional foundations for students.

Courses	CPMK (Course Objectives)	Relevansi CPL	SKS Teori	SKS Praktek	Type
Economic Survival / Internship Experience for Computer Science	Mampu menganalisa potensi teknologi sistem informasi dan merancang solusi berbasis teknologi Sistem Informasi	CPL1, CPL2, CPL3, CPL4, CPL5, CPL6, CPL7, CPL8	2	1	Mata Kuliah Universitas
Matematika Diskrit (<i>Discrete Mathematic</i>)	Mahasiswa mampu menggunakan konsep matematika diskrit (logika, himpunan, relasi/fungsi, kombinatorika, graf, dan teknik pembuktian) untuk memodelkan serta menganalisis masalah dasar computing, dan menyajikan solusi/pembuktian secara jelas dan benar.	CPL 1, CPL 3, CPL 4, CPL 8	3	0	Mata Kuliah Fakultas
Konsep Pemrograman (<i>Programming Concept</i>)	Mahasiswa mampu menganalisis permasalahan sederhana, merancang algoritma, mengimplementasikan program yang memenuhi requirement menggunakan tools modern, melakukan pengujian/debugging, serta mendokumentasikan dan mengkomunikasikan hasilnya secara profesional.	CPL1, CPL 2, CPL 3, CPL 4, CPL 5, CPL8	1	2	Mata Kuliah Fakultas
Perancangan Web (<i>Web Design</i>)	Mahasiswa mampu merancang dan membangun antarmuka web yang usable dan responsif berdasarkan kebutuhan pengguna, mengimplementasikannya dengan teknologi web dasar dan tools modern, mengevaluasi kualitasnya (UI/UX, aksesibilitas dasar), serta mempresentasikan keputusan desain secara jelas.	CPL1, CPL 2, CPL 3, CPL 5, CPL 6, CPL8	1	2	Mata Kuliah Fakultas
Pengantar Manajemen dan Organisasi (<i>Management and Organization</i>)	Mahasiswa mampu memahami dan menganalisis konsep manajemen serta organisasi dalam konteks kerja, berkomunikasi profesional, bekerja efektif dalam tim (termasuk kepemimpinan dasar), serta menunjukkan sikap etis dan pembelajaran mandiri untuk mendukung praktik sistem informasi di lingkungan industri/bisnis.	CPL3, CPL4, CPL5, CPL6, CPL7, CPL8	3	0	Mata Kuliah Prodi
Total			10	5	

Semester 2

Semester 2 strengthens system development and automation capabilities. Students begin to integrate programming, database, software design, business process automation, and entrepreneurial thinking in more applied learning activities.

Courses	CPMK (Course Objectives)	Relevansi CPL	SKS Teori	SKS Praktek	Type
Gen-AI & Economic Survival / Internship Experience for Computer Science	Mahasiswa mampu merancang dan memvalidasi gagasan bisnis/startup berbasis sistem informasi (problem–solution fit dan market fit), menyusun model bisnis dan strategi go-to-market, membuat prototipe/MVP sederhana dengan tools modern, mengevaluasi kelayakan dan risiko berbasis data (termasuk aspek legal dan etika), serta mengkomunikasikan proposal bisnis secara profesional melalui pitch dan dokumen pendukung.	CPL1, CPL2, CPL3, CPL4, CPL5, CPL6, CPL7, CPL8	1	2	Mata Kuliah Universitas
Pemrograman Aplikasi Web (<i>Web Application Development</i>)	Mahasiswa mampu merancang, membangun, dan menguji aplikasi web sesuai requirement dengan menerapkan prinsip pengembangan yang baik serta menggunakan tools modern, termasuk aspek keamanan dasar, dokumentasi, dan komunikasi hasil secara profesional (individu maupun tim).	CPL1, CPL2, CPL3, CPL4, CPL5, CPL6, CPL8	1	2	Mata Kuliah Fakultas
Sistem Database (<i>Database System</i>)	Mahasiswa mampu menganalisis kebutuhan data, merancang dan mengimplementasikan basis data relasional, menyusun query untuk kebutuhan operasional maupun analitik dasar, mengevaluasi kualitas solusi (integritas, performa, keamanan dasar), serta mendokumentasikan dan mempresentasikan hasilnya secara profesional.	CPL1, CPL2, CPL3, CPL4, CPL6, CPL8	1	2	Mata Kuliah Fakultas
Pemrograman Visual dan Berorientasi Objek (<i>Object Oriented and Visual Programming</i>)	Mahasiswa mampu merancang dan mengimplementasikan perangkat lunak berbasis paradigma berorientasi objek dengan antarmuka visual sesuai kebutuhan pengguna, melakukan pengujian dan evaluasi solusi menggunakan tools modern, serta menyajikan hasil pengembangan secara jelas dan profesional.	CPL1, CPL2, CPL3, CPL4, CPL5, CPL6, CPL8	1	2	Mata Kuliah Fakultas
Analisis dan Perancangan Proses Bisnis (<i>Business Process Analysis</i>)	Mahasiswa mampu mengidentifikasi, memodelkan, dan menganalisis proses bisnis pada studi kasus, menyusun rekomendasi perbaikan berbasis data/informasi yang relevan dengan konteks industri/bisnis, serta mengkomunikasikan hasil analisis secara profesional dalam kerja tim.	CPL1, CPL 2, CPL 3, CPL 4, CPL 5, CPL 6, CPL 7, CPL 8	3	0	Mata Kuliah Prodi
Total			7	8	

Semester 3

Semester 3 develops applied Information Systems competencies through mobile development, statistics, artificial intelligence, project management, security, UI/UX, and elective learning. This semester prepares students before entering concentration-based learning.

Courses	CPMK (Course Objectives)	Relevansi CPL	SKS Teori	SKS Praktek	Type
Pengembangan Aplikasi Mobile (<i>Mobile Application Development</i>)	Mahasiswa mampu merancang, mengembangkan, menguji, dan mendistribusikan aplikasi mobile sesuai requirement pengguna dengan memanfaatkan tools modern, memperhatikan aspek performa, usability, keamanan dasar, serta mendokumentasikan dan mengkomunikasikan hasil pengembangan secara profesional (individu maupun tim).	CPL1, CPL2, CPL3, CPL4, CPL5, CPL6, CPL8	1	2	Mata Kuliah Fakultas
Statistika Dan Probabilitas (<i>Probability and Statistics</i>)	Mahasiswa mampu menerapkan konsep probabilitas dan statistika untuk menganalisis data, melakukan inferensi dan pengujian hipotesis sederhana, mengevaluasi hasil analisis berbasis bukti, serta mengkomunikasikan interpretasi temuan secara jelas dalam konteks permasalahan sistem informasi/komputasi.	CPL1, CPL3, CPL4, CPL6, CPL8	1	2	Mata Kuliah Fakultas
Kecerdasan Artifisial untuk Sistem Informasi (<i>Artificial Intelligence for Information System</i>)	Mahasiswa mampu mengidentifikasi use-case AI pada sistem informasi, menyiapkan data, membangun dan mengevaluasi model AI/ML dasar menggunakan tools modern, serta mengintegrasikan hasilnya sebagai komponen solusi SI dengan mempertimbangkan etika, risiko, dan keterbatasan model.	CPL1, CPL2, CPL3, CPL4, CPL6, CPL8	1	2	Mata Kuliah Prodi
Perancangan Antarmuka dan Pengalaman Pengguna (<i>User Interface / User Experience Design</i>)	Mahasiswa mampu melakukan riset pengguna, merancang solusi UI/UX (user flow, wireframe, prototype) berdasarkan kebutuhan dan konteks bisnis, menguji dan mengevaluasi desain secara iteratif, serta mengkomunikasikan keputusan desain dan hasil evaluasi secara profesional dalam kerja tim.	CPL1, CPL2, CPL3, CPL5, CPL6, CPL8	1	2	Mata Kuliah Prodi
Infrastruktur Teknologi Informasi (<i>Information System Network & Infrastructure</i>)	Mahasiswa mampu menjelaskan konsep dasar jaringan dan infrastruktur SI, melakukan konfigurasi serta troubleshooting skenario sederhana menggunakan tools modern, menerapkan prinsip keamanan dan etika profesional, serta mendokumentasikan konfigurasi/hasil analisis secara jelas.	CPL1, CPL2, CPL3, CPL4, CPL6, CPL8	1	2	Mata Kuliah Prodi
Total			5	10	

Semester 4 (Concentration: Intelligent Enterprise System (ERP))

Semester 4 is the concentration boot camp semester. All students choose one of the concentration pathways: Intelligent Enterprise Systems OR Data Science for Business Analytics

Courses	CPMK (Course Objectives)	Relevansi CPL	SKS Teori	SKS Praktek	Type
Dasar-Dasar Perencanaan Sumber Daya Perusahaan (<i>Fundamental of Enterprise Resource Planning</i>)	Mahasiswa mampu memahami konsep dan arsitektur ERP serta alur proses lintas fungsi, melakukan konfigurasi dasar dan penggunaan modul ERP pada skenario bisnis, serta mengevaluasi manfaat dan risiko implementasi ERP dalam organisasi.	CPL1, CPL2, CPL3, CPL4, CPL6, CPL8	1	2	Mata Kuliah Konsentrasi
Sistem Manajemen Rantai Pasokan (<i>Supply Chain Management System</i>)	Mahasiswa mampu memodelkan proses supply chain dan mengoperasikan/ mengkonfigurasi modul SCM pada ERP untuk mendukung perencanaan, pengadaan, persediaan, dan distribusi, serta menganalisis kinerja proses menggunakan data dan laporan sistem.	CPL1, CPL2, CPL3, CPL4, CPL5, CPL6, CPL8	1	2	Mata Kuliah Konsentrasi
Perencanaan Sumber Daya Manufaktur (<i>Manufacturing Resource Planning</i>)	Mahasiswa mampu memahami dan menerapkan konsep MRP dalam ERP (BOM, routing, planning, scheduling dasar), menjalankan skenario perencanaan produksi, serta mengevaluasi dampaknya terhadap biaya, persediaan, dan ketepatan pemenuhan permintaan.	CPL1, CPL2, CPL3, CPL4, CPL6, CPL8	1	2	Mata Kuliah Konsentrasi
Sistem Informasi Manajemen Penjualan (<i>Sales Management System</i>)	Mahasiswa mampu menjalankan proses end-to-end penjualan (lead/order sampai invoicing sesuai skenario), mengelola data pelanggan dan transaksi di ERP, menganalisis laporan penjualan, serta menyusun rekomendasi perbaikan proses penjualan berbasis data.	CPL1, CPL2, CPL3, CPL4, CPL6, CPL8	1	2	Mata Kuliah Konsentrasi

Sistem Pelaporan Keuangan dan Bisnis (<i>Finance and Business Reporting</i>)	Mahasiswa mampu mengoperasikan proses keuangan dasar pada ERP (GL, AP/AR, posting, closing sederhana) dan menghasilkan laporan keuangan/ manajerial, serta melakukan interpretasi laporan untuk mendukung pengambilan keputusan secara akuntabel dan etis.	CPL2, CPL3, CPL4, CPL6, CPL8	1	2	Mata Kuliah Konsentrasi
Total			5	10	

Semester 4 (Concentration: Data Science for Business Analytics)

Semester 4 is the concentration boot camp semester. All students choose one of the concentration pathways: Intelligent Enterprise Systems OR Data Science for Business Analytics

Courses	CPMK (Course Objectives)	Relevansi CPL	SKS Teori	SKS Praktek	Type
Pengantar Sains Data (<i>Fundamental of Data Science</i>)	Mahasiswa mampu memahami konsep, alur kerja, dan peran data science dalam organisasi, mengidentifikasi problem/use-case yang tepat, menyiapkan data dan metrik keberhasilan, serta mengkomunikasikan insight awal secara profesional dan etis.	CPL1, CPL3, CPL4, CPL6, CPL8	1	2	Mata Kuliah Konsentrasi
Pemrograman untuk Sains Data (<i>Data Science Programming</i>)	Mahasiswa mampu mengolah data end-to-end menggunakan pemrograman dan tools modern (data wrangling, cleaning, exploratory analysis, pipeline sederhana), menerapkan praktik yang baik (reproducibility, dokumentasi), serta menyajikan hasil analisis secara jelas.	CPL1, CPL2, CPL3, CPL4, CPL6, CPL8	1	2	Mata Kuliah Konsentrasi

Pembelajaran Mesin (<i>Machine Learning</i>)	Mahasiswa mampu membangun, melatih, dan mengevaluasi model machine learning dasar sesuai problem domain, melakukan feature engineering dan validasi, menginterpretasi hasil berbasis data, serta mempertimbangkan risiko, bias, dan etika dalam penerapan model.	CPL1, CPL2, CPL3, CPL4, CPL6, CPL8	1	2	Mata Kuliah Konsentrasi
Penambangan Data (<i>Data Mining</i>)	Mahasiswa mampu menerapkan teknik data mining (mis. clustering, association rules, klasifikasi/regresi sesuai konteks) untuk menemukan pola/pengetahuan dari data organisasi, mengevaluasi kualitas temuan, dan menerjemahkannya menjadi rekomendasi yang bernilai untuk bisnis.	CPL1, CPL2, CPL3, CPL4, CPL6, CPL8	1	2	Mata Kuliah Konsentrasi
Visualisasi dan Pelaporan Data (<i>Data Visualization and Reporting</i>)	Mahasiswa mampu merancang visualisasi dan laporan data yang efektif untuk pengambilan keputusan, membangun dashboard/ reporting menggunakan tools modern, mengevaluasi kejelasan dan akurasi penyajian, serta mengkomunikasikan insight dan rekomendasi secara profesional.	CPL2, CPL3, CPL4, CPL6, CPL8	1	2	Mata Kuliah Konsentrasi
Total			5	10	

Semester 5

Courses	CPMK (Course Objectives)	Relevansi CPL	SKS Teori	SKS Praktek	Type
Analisis dan Desain Sistem Informasi (<i>Information System Analysis and Design</i>)	Mahasiswa mampu menganalisis kebutuhan sistem informasi dalam konteks organisasi/industri, memodelkan proses dan kebutuhan (fungsional/non-fungsional), merancang solusi SI yang sesuai requirement, serta mengkomunikasikan rancangan dan justifikasi desain secara profesional dalam kerja tim.	CPL1, CPL2, CPL3, CPL4, CPL5, CPL6, CPL8	1	2	Mata Kuliah Prodi

Manajemen Proyek Agile Sistem Informasi dengan DevOps (<i>Information System Agile Project Management with DevOps</i>)	Mahasiswa mampu merencanakan dan mengelola proyek pengembangan sistem informasi menggunakan pendekatan Agile, serta menerapkan praktik DevOps (versi kontrol, CI/CD dasar, kolaborasi, monitoring) untuk menghasilkan solusi yang memenuhi requirement, dievaluasi berbasis data, dan dikerjakan efektif dalam tim.	CPL2, CPL3, CPL4, CPL5, CPL6, CPL8	1	2	Mata Kuliah Prodi
Automasi Proses Bisnis (<i>Business Process Automation</i>)	Mahasiswa mampu mengidentifikasi proses bisnis yang layak di automasi, merancang dan mengimplementasikan otomasi menggunakan tools modern sesuai kebutuhan organisasi, mengevaluasi dampak otomasi berbasis data (efisiensi, risiko, kontrol), serta mendokumentasikan dan mempresentasikan solusi secara profesional dalam tim.	CPL1, CPL2, CPL3, CPL4, CPL5, CPL6, CPL7, CPL8	1	2	Mata Kuliah Prodi
Proyek Perencanaan Sumber Daya Perusahaan / Proyek Sains Data (<i>ERP Project / Data Science Project</i>)	Mahasiswa mampu melaksanakan tahapan proyek terkait dengan konsentrasi dengan kasus sebenarnya dari industri atau dunia usaha.	CPL1, CPL2, CPL3, CPL4, CPL5, CPL6, CPL7, CPL8	0	6	Mata Kuliah Konsentrasi
Total			3	12	

Semester 6

Courses	CPMK (Course Objectives)	Relevansi CPL	SKS Teori	SKS Praktek	Type
Magang 1 (<i>Internship 1</i>)	Mahasiswa mampu menunjukkan sikap profesional, disiplin, tanggung jawab, dan etika kerja, serta mampu menerapkan pengetahuan dan keterampilan di bidang Sistem Informasi dalam konteks organisasi atau industri nyata	CPL1, CPL2, CPL3, CPL4, CPL5, CPL6, CPL7, CPL8	0	15	Mata Kuliah Universitas
Total			0	15	

Semester 7

Courses	CPMK (Course Objectives)	Relevansi CPL	SKS Teori	SKS Praktek	Type
Magang 2 (<i>Internship 2</i>)	Mahasiswa mampu menunjukkan sikap profesional, disiplin, tanggung jawab, dan etika kerja, serta mampu menerapkan pengetahuan dan keterampilan di bidang Sistem Informasi dalam konteks organisasi atau industri nyata	CPL1, CPL2, CPL3, CPL4, CPL5, CPL6, CPL7, CPL8	0	15	Mata Kuliah Universitas
Total			0	15	

Semester 8

Courses	CPMK (Course Objectives)	Relevansi CPL	SKS Teori	SKS Praktek	Type
Capstone Project	Mahasiswa mampu mengidentifikasi masalah nyata pada sebuah industri, serta melakukan analisa, merancang, dan mengembangkan, mengintegrasikan, dan mengevaluasi solusi dalam lingkup manajemen proyek dan kerja sama tim yang baik	CPL1, CPL2, CPL3, CPL4, CPL5, CPL6, CPL7, CPL8	0	6	Mata Kuliah Universitas
Sistem Kecerdasan Bisnis dan Analitika Data (<i>Business Intelligence and Analytics</i>)	Mahasiswa mampu merancang dan membangun solusi Business Intelligence end-to-end (pengumpulan/transformasi data, data model/warehouse atau data mart sederhana, dashboard & reporting), menganalisis KPI untuk menghasilkan insight yang actionable, serta mengkomunikasikan rekomendasi keputusan secara profesional, etis, dan berbasis data.	CPL1, CPL2, CPL3, CPL4, CPL5, CPL6, CPL8	1	2	Mata Kuliah Prodi
Keamanan dan Penjaminan Sistem Informasi (<i>Information System Assurance and Security</i>)	Mahasiswa mampu mengidentifikasi risiko keamanan dan kebutuhan assurance pada sistem informasi, merancang kontrol keamanan dasar, melakukan penilaian dan pengujian sederhana (mis. risk assessment, vulnerability scanning dasar), serta menyusun rekomendasi mitigasi yang sesuai norma hukum dan etika.	CPL1, CPL2, CPL3, CPL4, CPL6, CPL8	1	2	Mata Kuliah Prodi
Sistem Informasi Korporat (<i>Enterprise Information System</i>)	Mahasiswa mampu menganalisis kebutuhan dan arsitektur sistem informasi enterprise, memahami integrasi data/proses lintas fungsi (termasuk ERP dan sistem analitik), merancang solusi konseptual yang sesuai requirement bisnis, serta mengkomunikasikan rancangan dan justifikasinya secara profesional dan etis.	CPL2, CPL3, CPL4, CPL6, CPL8	1	2	Mata Kuliah Prodi
Total			3	12	

Semester 9

Courses	CPMK (Course Objectives)	Relevansi CPL	SKS Teori	SKS Praktek	Type
Pancasila	Mahasiswa mampu memahami, menginternalisasi, dan menerapkan nilai-nilai Pancasila dalam pengambilan keputusan dan perilaku sebagai warga negara dan calon profesional, termasuk dalam konteks akademik/pekerjaan (kolaborasi, tanggung jawab, dan etika).	CPL 4, CPL 5, CPL 8	2	0	Mata Kuliah Umum
Agama	Mahasiswa mampu menginternalisasi nilai-nilai keagamaan untuk membangun karakter, integritas, dan tanggung jawab moral, serta menerapkannya dalam interaksi sosial dan praktik profesional secara menghargai keberagaman.	CPL 4, CPL 5, CPL 8	2	0	Mata Kuliah Umum
Kewarganegaraan	Mahasiswa mampu memahami hak dan kewajiban kewarganegaraan serta prinsip UUD 1945 dan tata kelola publik secara umum, lalu menerapkannya dalam sikap warga negara yang bertanggung jawab, kritis, dan etis dalam kehidupan bermasyarakat maupun konteks profesional.	CPL 4, CPL 5, CPL 8	2	0	Mata Kuliah Umum
Bahasa Indonesia	Mahasiswa mampu menginternalisasi nilai-nilai keagamaan untuk membangun karakter, integritas, dan tanggung jawab moral, serta menerapkannya dalam interaksi sosial dan praktik profesional secara menghargai keberagaman.	CPL 4, CPL 5, CPL 8	3	0	Mata Kuliah Umum
Total			9	0	

6.3 Course-PLO Mapping

The Course-PLO mapping table shows the relationship between courses and Program Learning Outcomes. The mapping is used to ensure that every PLO is supported by appropriate learning activities and that each course contributes clearly to graduate competence development.

Code	Course	PLO-1	PLO-2	PLO-3	PLO-4	PLO-5	PLO-6	PLO-7	PLO-8
FAC101	Economic Survival / Internship Experience for Computer Science	✓	✓	✓	✓	✓	✓	✓	✓
FAC102	Discrete Mathematic	✓		✓	✓				✓
FAC103	Programming Concept	✓	✓	✓	✓	✓			✓
FAC104	Web Design	✓	✓	✓		✓	✓		✓
ISC101	Business Process Analysis	✓	✓	✓	✓	✓	✓	✓	✓
ISC102	Information System Network & Infrastructure	✓	✓	✓	✓		✓		✓
ISC103	Management and Organization			✓	✓	✓	✓	✓	✓
FAC201	Gen-AI & Economic Survival / Internship Experience for Computer Science	✓	✓	✓	✓	✓	✓	✓	✓
FAC202	Web Application Development	✓	✓	✓	✓	✓	✓		✓
FAC203	Database System	✓	✓	✓	✓		✓		✓
FAC204	Object Oriented and Visual Programming	✓	✓	✓	✓	✓	✓		✓
ISC201	Information System Analysis and Design	✓	✓	✓	✓	✓	✓		✓
ISC202	Business Process Automation	✓	✓	✓	✓	✓	✓	✓	✓
UNV301	Pancasila				✓	✓			✓
UNV302	Religion				✓	✓			✓
UNV303	Citizenship				✓	✓			✓
UNV304	Bahasa Indonesia			✓	✓				✓
FAC401	Mobile Application Development	✓	✓	✓	✓	✓	✓		✓
FAC402	Probability and Statistics	✓		✓	✓		✓		✓
ISC401	Artificial Intelligence for Information System	✓	✓	✓	✓		✓		✓
ISC402	Information System Agile Project Management with DevOps		✓	✓	✓	✓	✓		✓
ISC403	Information System Assurance and Security	✓	✓	✓	✓		✓		✓
ISC404	User Interface / User Experience Design	✓	✓	✓		✓	✓		✓
ISEL401	General Elective	✓	✓		✓		✓		✓
ISC501	Enterprise Information System		✓	✓	✓		✓		✓
DSB501	Fundamental of Data Science for Business Analytics	✓		✓	✓		✓		✓
DSB502	Data Science Programming	✓	✓	✓	✓		✓		✓
DSB503	Machine Learning for Business Analytics	✓	✓	✓	✓		✓		✓
DSB504	Data Mining for Business Insight	✓	✓	✓	✓		✓		✓
DSB505	Data Visualization and Business Reporting		✓	✓	✓		✓		✓

ISC601	Business Intelligence and Analytics	✓	✓	✓	✓	✓	✓		✓
DSB601	Data Science for Business Analytics Project	✓	✓	✓	✓	✓	✓		✓
IES501	Fundamental of Intelligent Enterprise Systems	✓	✓	✓	✓		✓		✓
IES502	Supply Chain Management System	✓	✓	✓	✓	✓	✓		✓
IES503	Sales Management System	✓	✓	✓	✓		✓		✓
IES504	Manufacturing Resource Planning	✓	✓	✓	✓		✓		✓
IES505	Finance Management System and Reporting		✓	✓	✓		✓		✓
IES601	Intelligent Enterprise Systems Project	✓	✓	✓	✓	✓	✓		✓
AEA501	Fundamental of AI-Powered Enterprise Automation	✓	✓	✓	✓		✓		✓
AEA502	Business Process Automation with RPA	✓	✓	✓	✓	✓	✓		✓
AEA503	AI Agents and Workflow Automation	✓	✓	✓	✓		✓		✓
AEA504	Low-Code Application and API Integration	✓	✓	✓	✓		✓		✓
AEA505	Intelligent Document Processing and Process Mining	✓	✓	✓	✓		✓		✓
AEA601	AI-Powered Enterprise Automation Project	✓	✓	✓	✓	✓	✓		✓
WIL701	Career Stream in Information System	✓	✓	✓	✓	✓	✓	✓	✓
WIL801	Career Stream in Information System	✓	✓	✓	✓	✓	✓	✓	✓
CAP801	Pre Capstone Project	✓	✓	✓	✓	✓	✓		✓
CAP901	Capstone Project	✓	✓	✓	✓	✓	✓		✓

7. Course Details

FAC101 - Economic Survival / Internship Experience for Computer Science

Component	Description
Course Code	FAC101
Course Name	Economic Survival / Internship Experience for Computer Science
Semester	1
Credits	2
Course Type	Faculty Compulsory
Concentration	All
Course Description	This course introduces students to economic survival, professional exposure, and early career awareness in computer science and information systems contexts.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-5, PLO-6, PLO-7, PLO-8
Course Objective / CPMK	Students are able to analyze the potential of information systems technology and design technology-based ideas that respond to practical needs.

FAC102 - Discrete Mathematic

Component	Description
Course Code	FAC102
Course Name	Discrete Mathematic
Semester	1
Credits	3
Course Type	Faculty Compulsory
Concentration	All
Course Description	This course covers logic, sets, relations, functions, combinatorics, graphs, and proof techniques as mathematical foundations for computing.
Related PLO	PLO-1, PLO-3, PLO-4, PLO-8
Course Objective / CPMK	Students are able to use discrete mathematics concepts to model and analyze basic computing problems and present solutions clearly and correctly.

FAC103 - Programming Concept

Component	Description
Course Code	FAC103
Course Name	Programming Concept
Semester	1
Credits	3
Course Type	Faculty Compulsory
Concentration	All
Course Description	This course introduces algorithmic thinking, basic programming constructs, problem decomposition, debugging, and documentation.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-5, PLO-8
Course Objective / CPMK	Students are able to analyze simple problems, design algorithms, implement programs using modern tools, test/debug the programs, and document the results professionally.

FAC104 - Web Design

Component	Description
Course Code	FAC104
Course Name	Web Design
Semester	1
Credits	3
Course Type	Faculty Compulsory
Concentration	All
Course Description	This course introduces web interface design, usability, responsive design, basic HTML/CSS/JavaScript, and introductory accessibility concepts.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-5, PLO-6, PLO-8
Course Objective / CPMK	Students are able to design and build usable and responsive web interfaces based on user needs and communicate design decisions clearly.

ISC101 - Business Process Analysis

Component	Description
Course Code	ISC101
Course Name	Business Process Analysis
Semester	1
Credits	3
Course Type	Study Program Core
Concentration	All
Course Description	This course introduces business process identification, process modeling, process analysis, and improvement recommendations for organizational contexts.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-5, PLO-6, PLO-7, PLO-8
Course Objective / CPMK	Students are able to identify, model, and analyze business processes and prepare improvement recommendations based on relevant data and industry context.

ISC102 - Information System Network & Infrastructure

Component	Description
Course Code	ISC102
Course Name	Information System Network & Infrastructure
Semester	1
Credits	3
Course Type	Study Program Core
Concentration	All
Course Description	This course covers basic networking, infrastructure components, configuration, troubleshooting, documentation, and basic security practices.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-6, PLO-8
Course Objective / CPMK	Students are able to explain, configure, troubleshoot, and document basic information systems network and infrastructure scenarios using modern tools.

ISC103 - Management and Organization

Component	Description
Course Code	ISC103
Course Name	Management and Organization
Semester	1
Credits	3

Course Type	Study Program Core
Concentration	All
Course Description	This course introduces management, organizational behavior, teamwork, leadership, and organizational context for Information Systems practice.
Related PLO	PLO-3, PLO-4, PLO-5, PLO-6, PLO-7, PLO-8
Course Objective / CPMK	Students are able to analyze management and organizational concepts, communicate professionally, work effectively in teams, and demonstrate ethical and independent learning attitudes.

FAC201 - Gen-AI & Economic Survival / Internship Experience for Computer Science

Component	Description
Course Code	FAC201
Course Name	Gen-AI & Economic Survival / Internship Experience for Computer Science
Semester	2
Credits	4
Course Type	Faculty Compulsory
Concentration	All
Course Description	This course strengthens entrepreneurial and professional readiness by integrating generative AI tools, business validation, prototyping, and pitching.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-5, PLO-6, PLO-7, PLO-8
Course Objective / CPMK	Students are able to design and validate IS-based business/start-up ideas, create simple MVPs using modern tools, assess risks, and communicate a professional proposal.

FAC202 - Web Application Development

Component	Description
Course Code	FAC202
Course Name	Web Application Development
Semester	2
Credits	3
Course Type	Faculty Compulsory
Concentration	All
Course Description	This course covers development of dynamic web applications, database interaction, authentication basics, testing, and deployment principles.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-5, PLO-6, PLO-8
Course Objective / CPMK	Students are able to design, build, test, document, and present web applications that meet requirements using modern development tools.

FAC203 - Database System

Component	Description
Course Code	FAC203
Course Name	Database System
Semester	2
Credits	3
Course Type	Faculty Compulsory
Concentration	All
Course Description	This course covers relational database design, SQL, data integrity, query formulation, and basic data management for operational and analytical needs.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-6, PLO-8
Course Objective / CPMK	Students are able to analyze data needs, design and implement relational databases, formulate queries, and evaluate database quality and security basics.

FAC204 - Object Oriented and Visual Programming

Component	Description
Course Code	FAC204
Course Name	Object Oriented and Visual Programming
Semester	2
Credits	3
Course Type	Faculty Compulsory
Concentration	All
Course Description	This course develops object-oriented programming and visual interface development skills for user-oriented software solutions.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-5, PLO-6, PLO-8

Course Objective / CPMK	Students are able to design and implement object-oriented software with visual interfaces, test the solutions, and present the results professionally.
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ISC201 - Information System Analysis and Design

Component	Description
Course Code	ISC201
Course Name	Information System Analysis and Design
Semester	2
Credits	3
Course Type	Study Program Core
Concentration	All
Course Description	This course covers requirements elicitation, system modeling, functional and non-functional requirements, process modeling, data flow, and system design.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-5, PLO-6, PLO-8
Course Objective / CPMK	Students are able to analyze organizational information system needs, model requirements, design IS solutions, and communicate design justification in teams.

ISC202 - Business Process Automation

Component	Description
Course Code	ISC202
Course Name	Business Process Automation
Semester	2
Credits	3
Course Type	Study Program Core
Concentration	All
Course Description	This course introduces business process automation, automation opportunity analysis, basic workflow automation, risk assessment, and impact evaluation.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-5, PLO-6, PLO-7, PLO-8
Course Objective / CPMK	Students are able to identify automatable business processes, design and implement automation using modern tools, evaluate impact, and present solutions professionally.

UNV301 - Pancasila

Component	Description
Course Code	UNV301
Course Name	Pancasila
Semester	3
Credits	2
Course Type	University Compulsory
Concentration	All
Course Description	This course introduces Pancasila values and their application in citizenship, professional behavior, ethics, and responsible decision-making.
Related PLO	PLO-4, PLO-5, PLO-8
Course Objective / CPMK	Students are able to internalize and apply Pancasila values in decision-making and behavior as responsible citizens and future professionals.

UNV302 - Religion

Component	Description
Course Code	UNV302
Course Name	Religion
Semester	3
Credits	2
Course Type	University Compulsory
Concentration	All
Course Description	This course strengthens religious values, integrity, moral responsibility, and respect for diversity in social and professional interactions.
Related PLO	PLO-4, PLO-5, PLO-8
Course Objective / CPMK	Students are able to internalize religious values to build character, integrity, and moral responsibility in social and professional contexts.

UNV303 - Citizenship

Component	Description
Course Code	UNV303

Course Name	Citizenship
Semester	3
Credits	2
Course Type	University Compulsory
Concentration	All
Course Description	This course covers citizenship rights and responsibilities, constitutional principles, public governance, and ethical participation in society.
Related PLO	PLO-4, PLO-5, PLO-8
Course Objective / CPMK	Students are able to apply citizenship principles and responsible, critical, and ethical attitudes in society and professional contexts.

UNV304 - Bahasa Indonesia

Component	Description
Course Code	UNV304
Course Name	Bahasa Indonesia
Semester	3
Credits	3
Course Type	University Compulsory
Concentration	All
Course Description	This course develops academic Indonesian communication, structured writing, argumentation, citation, and professional communication skills.
Related PLO	PLO-3, PLO-4, PLO-8
Course Objective / CPMK	Students are able to communicate ideas in Indonesian clearly, logically, ethically, and appropriately in academic and professional contexts.

FAC401 - Mobile Application Development

Component	Description
Course Code	FAC401
Course Name	Mobile Application Development
Semester	4
Credits	3
Course Type	Faculty Compulsory
Concentration	All
Course Description	This course covers mobile application design, development, testing, distribution, performance, usability, and basic security.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-5, PLO-6, PLO-8
Course Objective / CPMK	Students are able to design, develop, test, and distribute mobile applications based on user requirements using modern tools.

FAC402 - Probability and Statistics

Component	Description
Course Code	FAC402
Course Name	Probability and Statistics
Semester	4
Credits	3
Course Type	Faculty Compulsory
Concentration	All
Course Description	This course covers probability, descriptive statistics, inference, hypothesis testing, and interpretation for computing and IS problems.
Related PLO	PLO-1, PLO-3, PLO-4, PLO-6, PLO-8
Course Objective / CPMK	Students are able to apply probability and statistics to analyze data, evaluate evidence, and communicate findings in Information Systems contexts.

ISC401 - Artificial Intelligence for Information System

Component	Description
Course Code	ISC401
Course Name	Artificial Intelligence for Information System
Semester	4
Credits	3
Course Type	Study Program Core
Concentration	All
Course Description	This course introduces AI use cases in Information Systems, data preparation, basic AI/ML modeling, evaluation, integration, and responsible AI considerations.

Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-6, PLO-8
Course Objective / CPMK	Students are able to identify AI use cases, prepare data, build and evaluate basic AI/ML models, and integrate AI as part of an IS solution responsibly.

ISC402 - Information System Agile Project Management with DevOps

Component	Description
Course Code	ISC402
Course Name	Information System Agile Project Management with DevOps
Semester	4
Credits	3
Course Type	Study Program Core
Concentration	All
Course Description	This course covers Agile planning, teamwork, version control, basic CI/CD, monitoring, and DevOps collaboration practices in IS projects.
Related PLO	PLO-2, PLO-3, PLO-4, PLO-5, PLO-6, PLO-8
Course Objective / CPMK	Students are able to plan and manage IS development projects using Agile and basic DevOps practices to deliver evaluated solutions in teams.

ISC403 - Information System Assurance and Security

Component	Description
Course Code	ISC403
Course Name	Information System Assurance and Security
Semester	4
Credits	3
Course Type	Study Program Core
Concentration	All
Course Description	This course introduces IS security risks, assurance concepts, control design, basic risk assessment, vulnerability assessment, and ethical/legal considerations.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-6, PLO-8
Course Objective / CPMK	Students are able to identify security and assurance needs, design basic controls, conduct basic assessments, and recommend ethical and legally compliant mitigation.

ISC404 - User Interface / User Experience Design

Component	Description
Course Code	ISC404
Course Name	User Interface / User Experience Design
Semester	4
Credits	3
Course Type	Study Program Core
Concentration	All
Course Description	This course covers user research, user flow, wireframing, prototyping, usability testing, and iterative UI/UX evaluation.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-5, PLO-6, PLO-8
Course Objective / CPMK	Students are able to conduct user research, design and test UI/UX prototypes, and communicate design decisions in team-based projects.

ISEL401 - General Elective

Component	Description
Course Code	ISEL401
Course Name	General Elective
Semester	4
Credits	3
Course Type	Study Program Elective
Concentration	All
Course Description	Students select one elective course such as Geographic Information System, Accounting Information System, or Government Information System to broaden applied IS competencies.
Related PLO	PLO-1, PLO-2, PLO-4, PLO-6, PLO-8
Course Objective / CPMK	Students are able to apply Information Systems concepts to a selected domain and produce an appropriate analysis or solution proposal.

ISC501 - Enterprise Information System

Component	Description
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Course Code	ISC501
Course Name	Enterprise Information System
Semester	5
Credits	3
Course Type	Study Program Core
Concentration	All
Course Description	This course introduces enterprise IS architecture, cross-functional process/data integration, ERP and analytics concepts, and enterprise solution design.
Related PLO	PLO-2, PLO-3, PLO-4, PLO-6, PLO-8
Course Objective / CPMK	Students are able to analyze enterprise information system needs and architecture, design conceptual solutions, and communicate design justifications professionally and ethically.

DSB501 - Fundamental of Data Science for Business Analytics

Component	Description
Course Code	DSB501
Course Name	Fundamental of Data Science for Business Analytics
Semester	5
Credits	3
Course Type	Concentration Bootcamp
Concentration	Data Science for Business Analytics
Course Description	This course introduces data science concepts, workflows, organizational data use cases, metrics, and ethical communication of insights.
Related PLO	PLO-1, PLO-3, PLO-4, PLO-6, PLO-8
Course Objective / CPMK	Students are able to explain data science workflows, identify business analytics problems, define data requirements and success metrics, and communicate initial insights.

DSB502 - Data Science Programming

Component	Description
Course Code	DSB502
Course Name	Data Science Programming
Semester	5
Credits	3
Course Type	Concentration Bootcamp
Concentration	Data Science for Business Analytics
Course Description	This course develops end-to-end data programming skills including wrangling, cleaning, exploratory analysis, reproducibility, and simple pipelines.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-6, PLO-8
Course Objective / CPMK	Students are able to process data end-to-end using modern programming tools and present clear and reproducible analytical outputs.

DSB503 - Machine Learning for Business Analytics

Component	Description
Course Code	DSB503
Course Name	Machine Learning for Business Analytics
Semester	5
Credits	3
Course Type	Concentration Bootcamp
Concentration	Data Science for Business Analytics
Course Description	This course covers basic machine learning workflow, feature engineering, model training, validation, interpretation, bias, and ethical issues.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-6, PLO-8
Course Objective / CPMK	Students are able to build, train, evaluate, and interpret basic machine learning models for business analytics problems responsibly.

DSB504 - Data Mining for Business Insight

Component	Description
Course Code	DSB504
Course Name	Data Mining for Business Insight
Semester	5
Credits	3
Course Type	Concentration Bootcamp
Concentration	Data Science for Business Analytics

Course Description	This course covers data mining techniques such as clustering, association rules, classification, and regression for discovering organizational patterns.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-6, PLO-8
Course Objective / CPMK	Students are able to apply data mining techniques to discover patterns, evaluate findings, and translate them into business recommendations.

DSB505 - Data Visualization and Business Reporting

Component	Description
Course Code	DSB505
Course Name	Data Visualization and Business Reporting
Semester	5
Credits	3
Course Type	Concentration Bootcamp
Concentration	Data Science for Business Analytics
Course Description	This course covers data storytelling, dashboard design, reporting tools, KPI presentation, and visual communication for decision-making.
Related PLO	PLO-2, PLO-3, PLO-4, PLO-6, PLO-8
Course Objective / CPMK	Students are able to design dashboards and reports that communicate insights and recommendations clearly, accurately, and ethically.

ISC601 - Business Intelligence and Analytics

Component	Description
Course Code	ISC601
Course Name	Business Intelligence and Analytics
Semester	6
Credits	3
Course Type	Study Program Core
Concentration	All
Course Description	This course covers BI end-to-end: data collection, transformation, data model/data mart, dashboarding, KPI analysis, and decision recommendations.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-5, PLO-6, PLO-8
Course Objective / CPMK	Students are able to design and build BI solutions, analyze KPIs, and communicate actionable recommendations based on data.

DSB601 - Data Science for Business Analytics Project

Component	Description
Course Code	DSB601
Course Name	Data Science for Business Analytics Project
Semester	6
Credits	6
Course Type	Concentration Project
Concentration	Data Science for Business Analytics
Course Description	This project course requires students to deliver a comprehensive data science solution for a business or organizational problem.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-5, PLO-6, PLO-8
Course Objective / CPMK	Students are able to complete a data science project from problem framing to evaluation and reporting, while working effectively in teams.

IES501 - Fundamental of Intelligent Enterprise Systems

Component	Description
Course Code	IES501
Course Name	Fundamental of Intelligent Enterprise Systems
Semester	5
Credits	3
Course Type	Concentration Bootcamp
Concentration	Intelligent Enterprise Systems
Course Description	This course introduces intelligent enterprise systems, ERP concepts, enterprise integration, AI-enabled enterprise functions, and implementation risks.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-6, PLO-8
Course Objective / CPMK	Students are able to explain intelligent enterprise system concepts, conduct basic ERP configuration/use scenarios, and evaluate benefits and risks.

IES502 - Supply Chain Management System

Component	Description
Course Code	IES502
Course Name	Supply Chain Management System
Semester	5
Credits	3
Course Type	Concentration Bootcamp
Concentration	Intelligent Enterprise Systems
Course Description	This course covers supply chain process modeling and ERP-supported procurement, inventory, distribution, reporting, and performance analysis.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-5, PLO-6, PLO-8
Course Objective / CPMK	Students are able to model supply chain processes, operate/configure ERP-supported SCM scenarios, and analyze process performance using system data.

IES503 - Sales Management System

Component	Description
Course Code	IES503
Course Name	Sales Management System
Semester	5
Credits	3
Course Type	Concentration Bootcamp
Concentration	Intelligent Enterprise Systems
Course Description	This course covers end-to-end sales processes, customer and transaction data, ERP sales modules, invoicing, and sales reporting.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-6, PLO-8
Course Objective / CPMK	Students are able to run end-to-end sales scenarios in enterprise systems, analyze sales reports, and recommend process improvements.

IES504 - Manufacturing Resource Planning

Component	Description
Course Code	IES504
Course Name	Manufacturing Resource Planning
Semester	5
Credits	3
Course Type	Concentration Bootcamp
Concentration	Intelligent Enterprise Systems
Course Description	This course covers MRP, bill of materials, routing, production planning, scheduling, and ERP-supported manufacturing decisions.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-6, PLO-8
Course Objective / CPMK	Students are able to apply MRP concepts in ERP scenarios and evaluate impacts on cost, inventory, and fulfillment.

IES505 - Finance Management System and Reporting

Component	Description
Course Code	IES505
Course Name	Finance Management System and Reporting
Semester	5
Credits	3
Course Type	Concentration Bootcamp
Concentration	Intelligent Enterprise Systems
Course Description	This course covers ERP-supported financial processes, general ledger, accounts payable/receivable, posting, closing, and managerial reporting.
Related PLO	PLO-2, PLO-3, PLO-4, PLO-6, PLO-8
Course Objective / CPMK	Students are able to operate basic ERP finance processes, generate financial/managerial reports, and interpret them for accountable decision-making.

IES601 - Intelligent Enterprise Systems Project

Component	Description
Course Code	IES601
Course Name	Intelligent Enterprise Systems Project
Semester	6
Credits	6

Course Type	Concentration Project
Concentration	Intelligent Enterprise Systems
Course Description	This project course requires students to deliver an enterprise systems implementation/configuration project based on a business case.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-5, PLO-6, PLO-8
Course Objective / CPMK	Students are able to design and execute an enterprise systems project, including requirements, configuration, testing, documentation, and team presentation.

AEA501 - Fundamental of AI-Powered Enterprise Automation

Component	Description
Course Code	AEA501
Course Name	Fundamental of AI-Powered Enterprise Automation
Semester	5
Credits	3
Course Type	Concentration Bootcamp
Concentration	AI-Powered Enterprise Automation
Course Description	This course introduces enterprise automation concepts, AI-enabled automation, workflow design, RPA, process governance, and automation opportunity assessment.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-6, PLO-8
Course Objective / CPMK	Students are able to identify enterprise automation opportunities, evaluate risks and benefits, and design initial AI-powered automation solutions.

AEA502 - Business Process Automation with RPA

Component	Description
Course Code	AEA502
Course Name	Business Process Automation with RPA
Semester	5
Credits	3
Course Type	Concentration Bootcamp
Concentration	AI-Powered Enterprise Automation
Course Description	This course develops RPA skills for automating repetitive business processes, including bot design, exception handling, testing, and documentation.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-5, PLO-6, PLO-8
Course Objective / CPMK	Students are able to design, implement, test, and document RPA-based automation for selected business processes.

AEA503 - AI Agents and Workflow Automation

Component	Description
Course Code	AEA503
Course Name	AI Agents and Workflow Automation
Semester	5
Credits	3
Course Type	Concentration Bootcamp
Concentration	AI-Powered Enterprise Automation
Course Description	This course covers AI agents, prompt/workflow design, human-in-the-loop controls, integration with enterprise workflows, and responsible automation.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-6, PLO-8
Course Objective / CPMK	Students are able to design AI-agent-assisted workflows, evaluate outputs and risks, and integrate automation into business processes responsibly.

AEA504 - Low-Code Application and API Integration

Component	Description
Course Code	AEA504
Course Name	Low-Code Application and API Integration
Semester	5
Credits	3
Course Type	Concentration Bootcamp
Concentration	AI-Powered Enterprise Automation
Course Description	This course covers low-code application development, API integration, workflow connectors, data exchange, and rapid enterprise solution prototyping.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-6, PLO-8
Course Objective / CPMK	Students are able to build low-code applications and API-integrated prototypes that support business process automation requirements.

AEA505 - Intelligent Document Processing and Process Mining

Component	Description
Course Code	AEA505
Course Name	Intelligent Document Processing and Process Mining
Semester	5
Credits	3
Course Type	Concentration Bootcamp
Concentration	AI-Powered Enterprise Automation
Course Description	This course covers document AI/OCR, information extraction, process mining concepts, event logs, process discovery, and automation improvement analysis.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-6, PLO-8
Course Objective / CPMK	Students are able to apply intelligent document processing and process mining techniques to identify automation opportunities and support evidence-based improvement.

AEA601 - AI-Powered Enterprise Automation Project

Component	Description
Course Code	AEA601
Course Name	AI-Powered Enterprise Automation Project
Semester	6
Credits	6
Course Type	Concentration Project
Concentration	AI-Powered Enterprise Automation
Course Description	This project course requires students to deliver an AI-powered automation solution for an enterprise or organizational process.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-5, PLO-6, PLO-8
Course Objective / CPMK	Students are able to complete an AI-powered automation project from problem analysis to implementation, evaluation, documentation, and team presentation.

WIL701 - Career Stream in Information System

Component	Description
Course Code	WIL701
Course Name	Career Stream in Information System
Semester	7
Credits	20
Course Type	University Compulsory
Concentration	All
Course Description	This course provides work-integrated or career-stream learning through internship, technopreneurship, research, mobility, or global networking pathways.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-5, PLO-6, PLO-7, PLO-8
Course Objective / CPMK	Students are able to apply IS and computing competencies in authentic career contexts and produce measurable outcomes through professional, ethical, and reflective practice.

WIL801 - Career Stream in Information System

Component	Description
Course Code	WIL801
Course Name	Career Stream in Information System
Semester	8
Credits	20
Course Type	University Compulsory
Concentration	All
Course Description	This course continues the career-stream pathway by deepening workplace, entrepreneurial, research, or global collaboration outputs.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-5, PLO-6, PLO-7, PLO-8
Course Objective / CPMK	Students are able to apply IS competencies to real activities in industry, entrepreneurship, research, or global collaboration and communicate outcomes reflectively.

CAP801 - Pre Capstone Project

Component	Description
Course Code	CAP801
Course Name	Pre Capstone Project
Semester	8
Credits	2

Course Type	University Compulsory
Concentration	All
Course Description	This course prepares students to define a real capstone problem, conduct literature or benchmarking study, elicit requirements, and design the proposal.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-5, PLO-6, PLO-8
Course Objective / CPMK	Students are able to identify and analyze a real problem, formulate scope and requirements, propose an initial solution design, and prepare a defensible capstone proposal.

CAP901 - Capstone Project

Component	Description
Course Code	CAP901
Course Name	Capstone Project
Semester	9
Credits	6
Course Type	University Compulsory
Concentration	All
Course Description	This course requires students to implement an Information Systems or computing solution based on an approved proposal and evaluate it in a relevant domain.
Related PLO	PLO-1, PLO-2, PLO-3, PLO-4, PLO-5, PLO-6, PLO-8
Course Objective / CPMK	Students are able to implement, test, and evaluate an IS/computing solution using modern tools and deliver a product, documentation, and professional presentation.

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- OECD. Artificial intelligence and the changing demand for skills in the labour market. OECD Artificial Intelligence Papers No. 14, 2024.
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